EXPLAINING A FEW-SHOT MEDICAL IMAGE SEGMENTER: A DETERMINISTIC GRAD-CAM AND GRADIENTSHAP STUDY OF UNIVERSEG

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Abstract

Deep segmentation networks deliver strong accuracy in medical imaging but remain opaque "black boxes," which is a critical obstacle in clinical settings where every prediction must be trusted and audited. This report integrates two complementary post-hoc Explainable AI (XAI) techniques, *Seg-Grad-CAM* (a gradient-based attribution method) and *GradientSHAP* (an expected-gradients approximation of Shapley values), into a few-shot medical image segmentation pipeline built on the pretrained UNIVERSEG model. The central methodological contribution is a *strictly deterministic* execution flow: a single seed locks all random number generators and `cuDNN`, the query, support set, and SHAP background distribution are extracted exactly once and frozen, and a binary prediction mask computed in a base forward pass is reused as the explanation target. This guarantees that the base prediction, the Grad-CAM heatmap, and the SHAP attribution all describe the *identical* model state, making the two explanations directly comparable. We evaluate qualitatively on two distinct modalities, ISIC skin-lesion dermoscopy and Kvasir-SEG colonoscopy polyps. Across both datasets, Grad-CAM consistently localizes the *macro* region driving the segmentation while GradientSHAP exposes the *micro*, pixel-level signed contributions concentrated on lesion texture and high-contrast boundaries. The agreement between the two methods, and their co-localization with the predicted foreground, provides qualitative evidence of model faithfulness, while disagreements expose over-segmentation failure modes. Our complete implementation and the deterministic XAI pipeline are publicly available at `https://github.com/canzorlu-eng/UniverSeg.git`.

1 Introduction

Modern computer vision models such as Convolutional Neural Networks (CNNs) and Vision Transformers achieve remarkable accuracy, yet their internal decision-making remains a *black box* (Adebayo et al., 2018). In high-stakes domains such as medical imaging, accuracy alone is insufficient: a clinician must be able to ask *where* the model looked and *why* a region was selected before a prediction can be trusted. Explainable AI (XAI) addresses this gap by transforming a model from a mere answer generator into an auditable, trustworthy partner.

This project studies XAI in the context of **few-shot medical image segmentation** using UNIVERSEG (Butoi et al., 2023), a model that segments a *query* image given a small *support set* of image–label pairs, without task-specific retraining. Segmentation poses a specific challenge for XAI: the model output is a dense $H \times W$ mask rather than a single class score, so attribution methods designed for classification must be adapted to produce a single differentiable target.

We integrate and contrast two methods that sit on the *gradient-based* branch of the XAI taxonomy (Section 2): Seg-Grad-CAM, which yields a coarse localization map, and GradientSHAP, which yields signed per-pixel attributions. Our contributions are:

1. **A strictly deterministic, reproducible XAI pipeline** that eliminates the seed/sampling drift which otherwise makes the base prediction, Grad-CAM, and SHAP describe slightly different model states and therefore incomparable explanations.

2. **A query-only, mask-targeted SHAP wrapper** that freezes the support set and reduces the segmentation output to a faithful scalar (foreground probability mass over the predicted lesion), avoiding gradient explosion and spatial-misalignment artifacts.
3. **A unified 1×5 visualization** and a cross-modality qualitative analysis on ISIC (skin lesions) and Kvasir-SEG (polyps), connecting the empirical heatmaps to the theory of attribution methods and to the faithfulness caveats of Adebayo et al. (2018).

2 Background and Related Work

A taxonomy of XAI for vision. Post-hoc explanation methods for vision models fall into several families. *Attribution / gradient-based methods*, exemplified by **Grad-CAM** (Selvaraju et al., 2017), use the gradients of a target concept flowing into a convolutional layer to produce a coarse localization heatmap; they are excellent for quickly debugging *where* a model looks and detecting background bias, but they explain *where* rather than *what* feature mattered. *Perturbation methods* such as **LIME** (Ribeiro et al., 2016) are model-agnostic: they partition an image into superpixels, randomly mask them, and measure the resulting drop in confidence; they are ideal for black-box APIs but require many forward passes. *Concept-based methods* such as **TCAV** (Kim et al., 2018) provide global explanations in terms of human-understandable concepts rather than pixel heatmaps. Finally, *intrinsically interpretable* (ante-hoc) models such as **ProtoPNet** (Chen et al., 2019) are interpretable by design, reasoning that "this part looks like that prototype."

Where our methods sit. We adopt two *gradient-based* explainers. Seg-Grad-CAM adapts Grad-CAM to dense prediction (Vinogradova et al., 2020). GradientSHAP, as implemented by the `shap` library's `GradientExplainer`, computes *expected gradients*, an extension of Integrated Gradients (Sundararajan et al., 2017) that approximates Shapley values (Lundberg & Lee, 2017). Importantly, GradientSHAP estimates the expectation of input gradients over a *background distribution* of reference images; it therefore inherits the baseline-centric philosophy of perturbation/game-theoretic methods while remaining computable from gradients. The two methods are thus complementary: one returns a smooth *macro* localization, the other a sparse, signed *micro* attribution.

The sanity-check problem. Adebayo et al. (2018) showed that visually pleasing saliency maps can be *misleading*: several popular methods produced near-identical, sensible looking heatmaps even after the trained weights were randomized, effectively behaving as edge detectors rather than reflecting what the network learned. The takeaway—explanations must be assessed for *faithfulness*, not just human visual appeal—directly motivates two of our design choices: (i) running two methodologically distinct explainers and checking their mutual agreement, and (ii) enforcing strict determinism so that any observed structure is attributable to the model state under test rather than to sampling noise.

Few-shot segmentation. UNIVERSEG (Butoi et al., 2023) performs in-context medical image segmentation: given a query image and a support set of image–label examples, it predicts the query mask through cross-image interaction blocks, without gradient updates at inference time. This conditioning on a frozen support set is precisely what our SHAP wrapper must hold fixed to isolate query-pixel attributions.

3 Methodology

3.1 Task and Model

UNIVERSEG takes three inputs—a query image $x_q \in \mathbb{R}^{1\times1\times H\times W}$, a support image tensor $x_s \in \mathbb{R}^{1\times S\times1\times H\times W}$, and support label masks y_s—and outputs query logits $z = f(x_q, x_s, y_s) \in \mathbb{R}^{1\times1\times H\times W}$. We operate on single-channel images resized to $H = W = 128$ with a support size $S = 3$. The foreground probability map is $p = \sigma(z)$.

3.2 A Strictly Deterministic, Reproducible Pipeline

The core requirement is that the base prediction, Grad-CAM, and SHAP all act on *one* fixed model state and *one* fixed set of inputs. Stochasticity otherwise enters through (a) PyTorch/CUDA RNG, (b) the shuffling `DataLoader`, and (c) random support/background sampling. We control all three:

- **Seed locking.** A single `set_seed(seed)` routine seeds `random`, `numpy`, `torch`, and CUDA, and sets `torch.backends.cudnn.deterministic = True` and `cudnn.benchmark = False`.
- **Dynamic but reproducible seeding.** By default the seed is drawn from `int(time.time())` so that different runs explore different images; a manual override (`TARGET_SEED`) reproduces any past run. The chosen seed is printed and appended to every output filename, e.g. `xai_panel_<image>_seed_<seed>.png`.
- **Frozen tensors.** The query, support images, and support masks are pulled from the loader *exactly once* and locked with `.detach().clone()` on the target device.
- **Static background.** The $N=8$ SHAP background queries are sampled *before* any forward pass and frozen as the baseline distribution.

The unified execution flow is: **(A)** a base forward pass yields the definitive probability map p and a frozen binary mask $m = \mathbf{1}[p > 0.5]$; **(B)** Grad-CAM is computed on the same frozen tensors; **(C)** SHAP is computed on the same frozen tensors and frozen background, targeting the same mask m.

3.3 Seg-Grad-CAM

We hook the last decoder convolution of UNIVERSEG. Because the output is a dense map, we reduce it to a scalar by averaging the logits over the predicted-positive pixels,

$$S_{\text{CAM}} = \frac{\sum_u z_u\, m_u}{\sum_u m_u}, \tag{1}$$

and backpropagate S_{CAM}. With activations A^k and gradients $\partial S_{\text{CAM}}/\partial A^k$, the channel weights are the global-average-pooled gradients α_k, and the map is

$$L_{\text{CAM}} = \text{ReLU}\left(\sum_k \alpha_k A^k\right), \tag{2}$$

bilinearly upsampled to 128×128 and min–max normalized to $[0, 1]$. Grad-CAM thus answers *where* the model looks.

3.4 GradientSHAP with a Query-Only, Mask-Targeted Wrapper

Two failure modes plague a naive application of `GradientExplainer` to a few-shot segmenter. First, the explainer expects a single perturbed input and a *scalar* output. Second, summing raw logits produces an enormous target that causes exploding gradients clustered in an image corner.

We resolve both with a wrapper module that (i) registers the support images, support masks, and the frozen lesion mask m as non-trainable buffers, so the explainer perturbs *only* the query, and (ii) defines the scalar target as the predicted foreground probability mass restricted to the lesion,

$$S_{\text{SHAP}}(x_q) \;=\; \sum_u \sigma\big(f(x_q, x_s, y_s)\big)_u\, m_u. \tag{3}$$

Using $\sigma(\cdot)$ bounds the target in $[0, 1]$ per pixel (no explosion), and masking by m focuses attribution on the pixels that actually formed the predicted lesion. `GradientExplainer` then estimates expected gradients of (3) with respect to the query, integrated over the frozen background distribution. The returned attribution tensor, which shares the $(\text{batch}, \text{channel}, H, W)$ layout of the input, is indexed explicitly to a 128×128 signed map; we never sum over spatial axes (which would collapse the map into a corner artifact). Positive (red) values indicate pixels that *increase* the predicted lesion probability; negative (blue) values indicate pixels that *suppress* it.

3.5 Unified Visualization

Each run emits a single 1×5 panel: *Query*, *Ground-Truth Mask*, *Base Prediction Mask*, *Grad-CAM Overlay* (jet colormap, $\alpha = 0.5$ over the query), and *SHAP Heatmap* (diverging `coolwarm`, symmetric about zero with a colorbar). Because of Section 3.2, all five sub-panels are guaranteed to describe one model state.

4 Experimental Setup

Datasets. We use two modalities with very different appearance statistics. **ISIC** (Codella et al., 2018) contains dermoscopic skin-lesion images; lesions are large, roughly elliptical, and often occluded by hair and ruler markings. **Kvasir-SEG** (Jha et al., 2020) contains colonoscopy frames with polyps; targets vary widely in size and share color with the surrounding mucosa, and frames include specular highlights and endoscope vignetting. All images are converted to grayscale and resized to 128×128; masks are binarized at 0.5.

Configuration. Pretrained UNIVERSEG; support size $S = 3$; SHAP background size $N = 8$; Grad-CAM target layer = last decoder convolution. The identical code path serves both datasets (`xai_unified.py` for ISIC, `xai_unified_kvasir.py` for Kvasir), differing only in data paths and the `dataset_type` flag.

Table 1: The two explainers occupy complementary roles in our pipeline.

	Seg-Grad-CAM	**GradientSHAP**
XAI family	Gradient attribution (CAM)	Expected gradients / Shapley
Granularity	Macro (coarse region)	Micro (per-pixel)
Sign	Non-negative (ReLU)	Signed ($\pm$)
Question answered	*Where* did it look?	*Which* pixels help/hurt?
Baseline needed	No	Yes (frozen background)
Output target	Mean logit over m	Prob. mass over m (Eq. 3)

5 Results and Analysis

We present two cases per dataset: a well-aligned "hero" case and a diagnostic case that exposes a failure mode. In every panel the reader should compare the *Base Prediction* (column 3) against the two explanations (columns 4–5).

5.1 ISIC: Skin-Lesion Dermoscopy

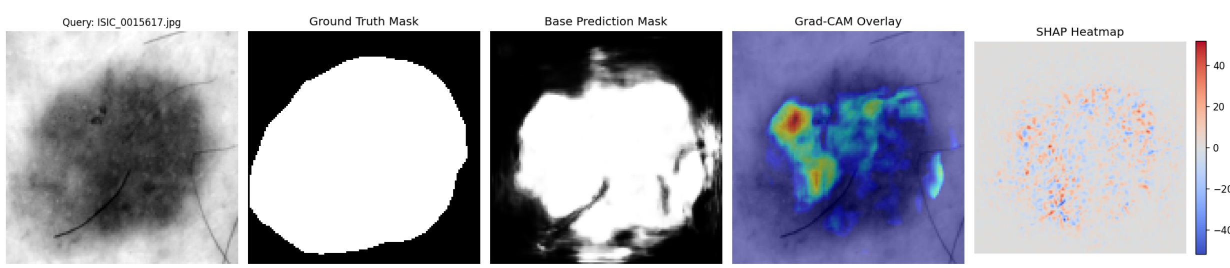

Figure 1: ISIC case `ISIC_0015617`. **Grad-CAM (macro):** a single contiguous warm blob covers the body of the pigmented lesion, confirming the model attends to the lesion interior rather than the surrounding skin or the thin hair artifacts. **GradientSHAP (micro):** a dense cloud of signed points tiles the same lesion territory; positive (red) responses concentrate on the darker, high-pigment texture that drives foreground probability up, while scattered blue points mark locally suppressive pixels. The two maps co-localize with the prediction, qualitative evidence of faithful attention.

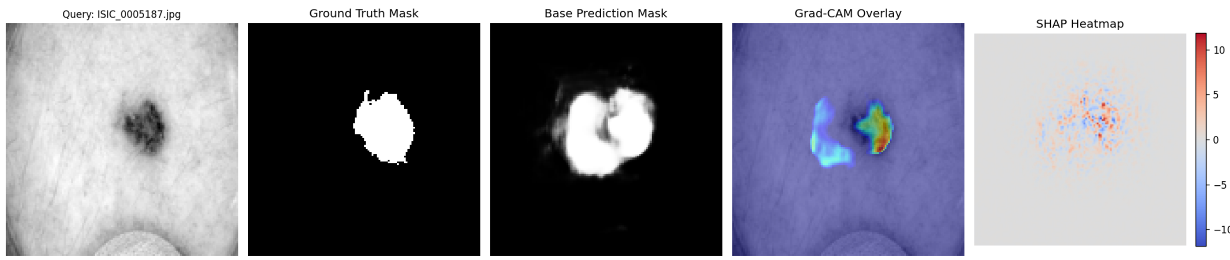

Figure 2: ISIC case `ISIC_0005187`, a small compact lesion. **Grad-CAM** fires as a tight, well-localized hot-spot exactly on the lesion, with a faint halo extending slightly left—mirroring the prediction's mild leftward over-extension beyond the ground truth. **GradientSHAP** forms a compact signed cluster on the lesion core and is essentially silent over the smooth healthy skin, demonstrating that the few high-contrast lesion pixels, not the uniform background, carry the attribution.

On ISIC, lesions are large and high-contrast, which is the ideal regime for both methods. Grad-CAM reliably paints the lesion body as one macro region (Fig. 1), while SHAP decomposes that region into the individual pigment-rich pixels responsible for the decision. Where the prediction slightly over-extends (Fig. 2), the Grad-CAM halo and the spatial spread of SHAP points extend in the *same* direction—the explanations faithfully track the model's error rather than the ground truth.

5.2 Kvasir-SEG: Colonoscopy Polyps

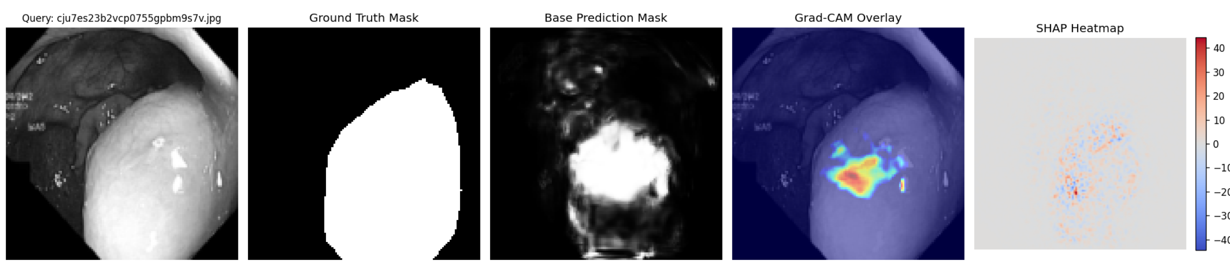

Figure 3: Kvasir case `cju7es...`. **Grad-CAM** concentrates a warm blob on the protruding polyp surface, discriminating it from the visually similar mucosal wall—a non-trivial success given that polyp and background share color. **GradientSHAP** clusters its strongest signed responses at the polyp centroid and along its illuminated boundary, indicating that contrast/illumination cues at the polyp surface are the micro-features pushing the foreground probability up.

Kvasir is the harder modality: polyps are camouflaged against the mucosa, and specular highlights and vignetting introduce strong non-lesion contrast. In the clean case (Fig. 3) both methods still localize the polyp, which is strong evidence that the model has learned a genuine polyp representation rather than a trivial brightness heuristic. The diagnostic case (Fig. 4) is equally valuable: when the model over-segments, the Grad-CAM hot-spot and the diffuse SHAP energy point to the bright mucosal folds that triggered the error, illustrating how XAI converts an opaque mistake into an actionable diagnosis.

5.3 Macro vs. Micro: A Cross-Dataset Synthesis

Three patterns hold across both modalities. **(1) Complementarity:** Grad-CAM answers *where* (one smooth region) and SHAP answers *which pixels and in which direction* (sparse signed points); neither alone tells the full story. **(2) Co-localization implies faithfulness:** when the prediction is correct, both maps land on the lesion and avoid the background, the agreement of two methodologically independent explainers being far more convincing than either in isolation. **(3) Disagreement localizes error:** when the prediction is wrong, the explanations move *with the model*, not with the ground truth, exposing the offending image features.

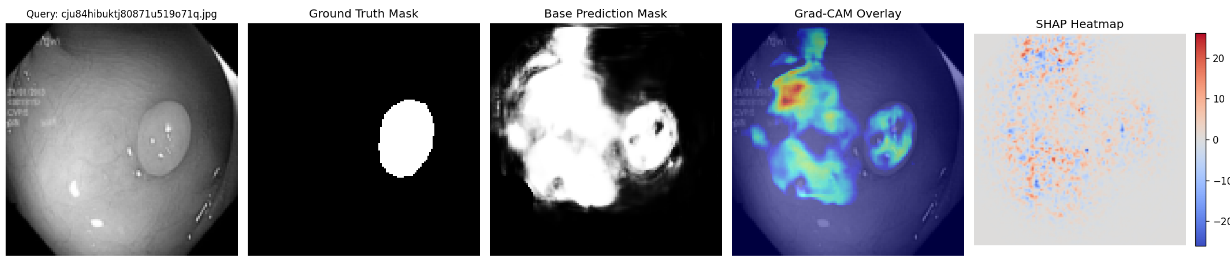

Figure 4: Kvasir case `cju84h...`, a *diagnostic* example. The ground-truth polyp (column 2) is small, yet the base prediction over-segments a large portion of the frame. **Grad-CAM** reveals *why*: its hot-spot sits above/left of the true polyp, on bright mucosal folds that the model has confused for lesion tissue. **GradientSHAP** spreads signed energy broadly rather than tightly, with mixed red/blue noise over the over-segmented area. Here the two explanations agree that the model is responding to spurious high-contrast mucosa—an interpretable account of a segmentation failure.

6 Discussion

Faithfulness, not just appeal. Following Adebayo et al. (2018), we treat visual plausibility as necessary but not sufficient. Our deterministic pipeline ensures that the heatmaps reflect a single, fixed model state, and the mutual agreement of Grad-CAM and GradientSHAP acts as an informal cross-method faithfulness check. The SHAP map also passes a basic input-sensitivity sanity check: it is near-zero over homogeneous background and active over informative texture, rather than acting as a uniform edge detector.

Limitations. The study is qualitative; a quantitative faithfulness protocol (e.g. model-parameter randomization, deletion/insertion curves) would strengthen the claims. GradientSHAP is sensitive to the choice of background distribution and to N; our $N = 8$ trades fidelity for the runtime of repeated forward passes through a few-shot model. Grad-CAM resolution is bounded by the decoder feature-map size. Finally, attributions are conditioned on the particular frozen support set, so explanations are statements about *this* in-context configuration of UNIVERSEG.

7 Conclusion

We built a strictly deterministic XAI pipeline that makes Grad-CAM and GradientSHAP directly comparable for a few-shot medical segmenter, and applied it across two modalities. Grad-CAM provides faithful macro localization and GradientSHAP provides signed micro attribution; their agreement supports model trust on correct cases and their behavior on incorrect cases turns silent errors into interpretable diagnoses. The result is a small but principled step toward auditable medical image segmentation.

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A Reproducibility

Every figure filename encodes its seed, e.g. `xai_panel_ISIC_0015617_seed_<seed>.png`. To reproduce a run, set `TARGET_SEED` to the printed value and re-run `python xai_unified.py` (ISIC) or `python xai_unified_kvasir.py` (Kvasir). The deterministic settings of Section 3.2 guarantee bit-for-bit identical base predictions, Grad-CAM maps, and SHAP attributions for a fixed seed. The full source code is available at `https://github.com/canzorlu-eng/UniverSeg.git`.